[PostgreSQL] SSL 적용

- SSL 적용 방법 가이드

서버 키 생성

```
[experdb2@single data]$ openssl genrsa -out server.key 2048
Generating RSA private key, 2048 bit long modulus (2 primes)
.....................................................................................................................................
.....................................................................+++++
e is 65537 (0x010001)
```

서버 인증서 서명 요청 생성

```
[experdb2@single data]$ openssl req -new -key server.key -out server.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:
State or Province Name (full name) []:
Locality Name (eg, city) [Default City]:
Organization Name (eg, company) [Default Company Ltd]:
Organizational Unit Name (eg, section) []:
Common Name (eg, your name or your server's hostname) []:
Email Address []:

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:
An optional company name []:
```

자체 서명된 서버 인증서 생성

```
[experdb2@single data]$ openssl x509 -req -days 3650 -in server.csr -signkey server.key -out server.crt
Signature ok
subject=C = XX, L = Default City, O = Default Company Ltd
Getting Private key
```

server.crt 복제

```
[experdb2@single data]$ cp server.crt root.crt
```

server.key 권한 변경

```
[experdb2@single data]$ chmod 600 server.key
```

postgresql.conf SSL 적용

```
[experdb2@single data]$ vi postgresql.conf
ssl = on
ssl_ca_file = '\\LOCATION_OF_FILE\\root.crt'
ssl_cert_file = '\\LOCATION_OF_FILE\\server.crt'
ssl_key_file = '\\LOCATION_OF_FILE\\server.key'
```

```
# ssl_ciphers = 'HIGH:MEDIUM:+3DES:!aNULL' # allowed SSL ciphers
# ssl_prefer_server_ciphers = on
```

pg_hba.conf SSL 적용

```
# TYPE  DATABASE        USER            ADDRESS                 METHOD

# "local" is for Unix domain socket connections only
local   all             all                                     trust
# IPv4 local connections:
hostssl    all             all             127.0.0.1/32            trust
# IPv6 local connections:
hostssl    all             all             ::1/128                 trust
# Allow replication connections from localhost, by a user with the
# replication privilege.
local   replication     all                                     trust
hostssl    replication     all             127.0.0.1/32            trust
hostssl    replication     all             ::1/128                 trust
hostssl    all             all                 0.0.0.0/0               md5
```

DB 재기동 후 접속 테스트

```
[experdb@single ~]$  psql -h 10.12.30.142 -U experdba -d experdb -p 15432
Password for user experdba:
psql (15.7)
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off)
Type "help" for help.

experdb=#
```